BILLERICA ENGINEERING AND DESIGN

Flying Leota

Project plan for a floating crystal-ball head: a four-line cable rig with an Ethernet ESP32 motion controller, driven by Show Controller over SCMP.

Owner: Nate Cory
Status: Research and planning
Date: November 2026

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Sources

A head that floats in the air, held by four motor-driven lines. The motion controller is an ESP32 with wired Ethernet and a local touchscreen. Show Controller controls it over the Show Controller Motion Protocol (SCMP).

Status: research and planning, November 2026. No code yet. Every "Proposed" item below needs a yes from Nate before it is built.

Summary

- **The prop:** a lifesize head inside an 18 in (0.46 m) clear crystal ball, likely 5–8 kg. It rests on a pedestal 1.0 m up, in the middle of an 8 ft × 8 ft tent. It rises about 0.68 m and drifts about ±0.69 m each way, always facing the audience. A projector, a Raspberry Pi Zero 2 W and a battery ride inside.
- **The rig:** four winches at the bases of four corner posts. Each line runs up its post, over a pulley at 8 ft and down to a small fitting on top of the ball, like Skycam. The same hardware flies in 3 axes, or in a fixed depth plane.
- **The controller:** an ESP32-P4 board with wired Ethernet and a 7 in touchscreen. Show Controller drives it over SCMP as three axes: x, y and z. The touchscreen shows status and has local controls: remote on/off, arm, home, park and stop.
- **The motors:** closed-loop steppers with fail-safe brakes are the lead choice. ODrive brushless servos are the alternative. One of each gets tested before buying four.
- **Safety:** two or more hardwired E-stops through a safety relay. A controlled stop, then power off, and the brakes hold the ball. The firmware keeps its own limits, tension checks and a network heartbeat. Software is never the only safety layer.
- **Weather:** a wind sensor plus line-tension monitoring. Green, amber and red levels. At red, the ball flies home to the dock and locks out until conditions settle.

- **Next:** buy the bench parts (about $800–1,100), then run two test tracks side by side: motion (one winch, both motor types, line samples) and the head (sphere, projector, power, weight).

1. Goals

#	Goal
G1	Move a head-sized prop in 3 axes (left/right, up/down, toward/away), like Skycam. A fixed-depth plane (2 axes) is an acceptable fallback.
G2	Show Controller drives it over Ethernet with SCMP. No Wi-Fi anywhere in the control path.
G3	A local touchscreen shows status. It also has controls to enable or disable remote control, arm and disarm, home, park and stop.
G4	At least two hardwired emergency stops. They work even if the software is dead.
G5	It runs outdoors under a tent, enclosed on 3 sides and seen from 1 side. It has clear wind rules and enforces them itself.

Non-goals for version 1: turning or tilting the head, and more than one flying object. The head's own projector and LEDs run on their own battery and are not part of the motion controller.

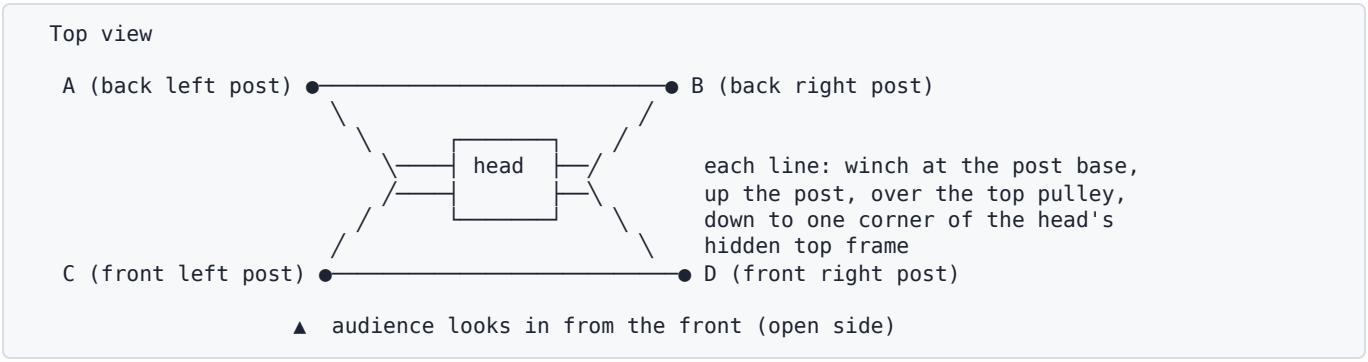
Site and prop (from Nate, 2026-11-03)

Item	Value
Tent	About 8 ft × 8 ft (2.44 m). Pulleys at about 8 ft (2.44 m). The roof peaks higher, but pulleys up there would sit closer to the center and cut the travel, so 8 ft it is.
Dock	A pedestal fixed to a table in the middle of the tent, its top 1.0 m off the floor. The head rests here when it isn't flying.
Head	Lifesize, inside a clear sphere made to look like a crystal ball, about 18 in (0.46 m) across. A large curly wig. Weight not known yet.
Facing	**The face must always point towards the audience.**
On board	A Raspberry Pi Zero 2 W drives a small wide-angle projector in the neck. A mirror throws the image onto the back of the face. Maybe LEDs. All on a lithium-ion battery. No power through the lines: they stay as thin as possible. Projector not chosen yet.
Viewing	At night, in low light. The lines should vanish into the dark.

This is like the séance-room crystal ball: it sits on the table, rises, drifts and floats, and settles back. That fits the space well, because the space is small (section 4).

2. How it works

Layout (proposed: Skycam style)



- Four posts at the corners of a rectangle. Each has a pulley at the top. Seen from above, the head is always inside that rectangle.
- Each line runs from a winch at the **base** of its post, up the post, over the top pulley, and down to the head. Skycam does the same thing. Heavy parts stay low and easy to reach, and the tops of the posts carry only pulleys.
- Gravity keeps the lines tight. The head hangs below the pulleys. It can't go above their plane, and it gets harder to hold as it nears that plane or a corner (section 4).
- The four lines attach to four points about 10 cm apart on a small fitting at the top of the ball (like the corners of Spidercam's camera dolly, but much closer together). If they all went to one point, the ball could spin. With four points, the face stays towards the audience. Section 4 explains the spacing.
- **The same hardware does both modes.** The 2-axis mode is a software setting: the depth axis is fixed at one value. Nothing is rebuilt to switch.

Why not the other layouts:

Layout	Axes	Problem for this prop
2 lines from the top (a "polargraph")	2	Nothing holds the depth. A breeze from the open side swings it like a pendulum.
2 lines from the top and 2 from the floor, in one plane	2	Stiff side to side, but depth is held only by line tension. That is the direction the wind blows through an open tent front. The floor lines are also in the scene.
8 lines (fully held)	3 + turning	Twice the hardware. Lines cross the scene from below.
4 lines from the top (Skycam)	**3, or 2**	**Recommended.** Front and back lines hold the depth directly.

The frame must not be the tent

Put the pulleys on their own freestanding frame inside the tent, not on the tent poles. Tent frames move in the wind, and the lines pull inward on whatever holds them. If the posts move, the head moves. At 8 ft, a frame of aluminum extrusion or steel strut channel works well: four posts, a rectangle tying their tops together, and wide ballasted feet. Section 4 gives the loads.

Lines

The goal is lines you can't see against a black curtain, that don't stretch, and that stay the same length all evening. Choose the line by testing (Phase 3). For now:

Line	Visibility	Stretch and creep	Notes
Nylon monofilament (clear)	Good at any color backdrop	High. Stretches noticeably under load, creeps, and **changes length when wet or humid**.	Needs constant re-trimming. Least suited to a damp outdoor tent.
Fluorocarbon (clear)	Good	Lower than nylon, still meaningful	Stiff, and has memory on a drum.
Braided UHMWPE (Dyneema/Spectra) fishing line, black, about 125–250 lb test (by head weight, section 4)	Thin (roughly 0.5–0.7 mm). Disappears against black.	Very low	**Recommended starting point.** Strong for its size. Spools well on a grooved drum.

Pick the line size from the loads in section 4. Theatre rigging commonly uses large design factors (often 8:1 to 10:1). Section 7 lists this as an open question.

3. Hardware recommendations

3.1 Controller: ESP32-P4

The ESP32-P4 is the only ESP32 that has an Ethernet MAC and a MIPI-DSI display interface together, with the speed to run a 7-inch UI and motion together.

Chip	Ethernet	Display	Verdict
ESP32 (original)	Built-in MAC, RMII PHY	SPI or parallel only. A 7-inch UI is slow.	Workable, but tight.
ESP32-S3	**No MAC.** Needs a W5500 over SPI.	RGB parallel works	Ethernet over SPI is slower and eats pins.
ESP32-P4	**Built-in MAC, RMII PHY**	**MIPI-DSI, 2D accelerator**	**Recommended.** Dual-core RISC-V at 360–400 MHz, 3 CAN (TWAI) controllers, 4 RMT channels (step pulses), and lots of GPIO.

The P4 has no radio, which suits the no-Wi-Fi rule. Most P4 boards add an ESP32-C6 for Wi-Fi. We leave it turned off.

Boards (all have Ethernet and a DSI connector):

Board	Screen	About	Use
Espressif ESP32-P4-Function-EV-Board	7" 1024×600 capacitive touch, in the kit	$55+ (board); more with the display kit	**Bench and first build.** Espressif's own board, with the best ESP-IDF board support (esp-bsp). IP101 Ethernet PHY.
Waveshare ESP32-P4-Module-DEV-KIT-B	7" DSI touch, in the kit	~$78 kit	Alternative. 40-pin GPIO header, optional PoE.
Olimex ESP32-P4-DevKit	Add a DSI panel	€16	Open-source hardware. Good for the final build on a custom I/O carrier board.
Waveshare ESP32-P4-ETH / -POE-ETH	Add a DSI panel	$25+	Small and simple, if we design our own carrier.

Waveshare's all-in-one P4 touch panels (4.3", 5", 7", 10.1") have **no Ethernet**, so they don't fit.

For the field build, plan a small **I/O carrier board** (or DIN-rail terminal breakouts) between the ESP32 and the outside world:

- Opto-isolated 24 V inputs: E-stop chain status, reset, drive alarms (×4), and the Local/Remote key.
- Differential step/dir outputs (26LS31 line driver) to the drives.
- 2 CAN transceivers (drives or I/O later), 1 RS-485 transceiver (anemometer).
- A watchdog "charge pump" output. The drives' enable line stays on only while the firmware keeps toggling a pin. A hung ESP32 drops the enable. Classic CNC practice.
- Load cell amplifier inputs (×4).

3.2 Winches: motors and drives

Load numbers come from section 4. The head's weight isn't known yet, so **size the winches for a 10 kg head**: up to 140 N on a line at the worst spot in the workspace. Lighter heads then just run with more margin. With a 12 mm radius drum (24 mm across), 140 N is 1.7 N·m. The space is small, so the head moves

slowly: 0.3 m/s line speed is about 240 rpm at the drum. A 3 N·m NEMA 23 does this directly, with no gearbox, because stepper torque is highest at low speed. Its brake must hold at least 2 N·m.

Sizing for 10 kg instead of 5 kg costs little on the motor side: the same motor class, a smaller drum, and maybe $100–200 more across four winches. The real cost of a heavy head is thicker line (section 4).

A hanging load needs brakes. If drive power drops (E-stop, power cut, fault), an unbraked motor lets the head fall. Use fail-safe brakes, which grip when power is removed. Removing drive power then engages the brakes too, which matches SCMP's "E-stop removes drive power" rule.

Option	Per winch	For	Against
A. Closed-loop stepper, NEMA 23, 2–3 N·m, with a 24 V fail-safe brake. Drive: StepperOnline CL57T V4.1 class, or Leadshine CS-D507E with the CS-M22313B brake motor. 48 V DC. Step/dir input, alarm output.	~$150–250	**Recommended for version 1.** Simple, cheap, high holding torque, brakes available. The encoder catches missed steps, and the drive raises an alarm on following error.	No direct torque control, so line tension is managed through load cells (section 4).
B. Brushless servo: ODrive S1 + M8325s motor, controlled over CAN, plus a separate brake.	~$300–350 + brake	Torque mode controls line tension directly. Motor current gives a tension reading. Hangprinter's cable printer is built this way.	More tuning. Brake and drum need custom mounts. Often out of stock.
C. Teknic ClearPath-SDSK integrated servo	~$385–600	Excellent, all in one, step/dir.	Expensive, and no brake option was found.
D. MKS SERVO57D (CAN or RS-485 closed-loop stepper)	~$50–80	Cheap. Good for bench experiments.	Thin documentation. Unproven for a hanging load.

Recommendation: **build version 1 on Option A, and write the firmware with a drive interface** so Option B can replace it later without changing the rest. One caveat: noise. The audience is close, at night, in a quiet scene, and steppers hum or whine. Buy one of each for Phase 3 (a single winch) and listen to both under load before buying four.

Closed-loop stepper or ODrive: the difference

Both are a motor, an encoder that measures where the shaft is, and a drive that powers the motor. They differ in **what you tell the drive**.

Closed-loop stepper: "be at this position." The ESP32 sends step pulses. Each pulse means "turn one tiny fixed angle more." The drive pushes as hard as it needs to keep the shaft exactly where the pulses say. The encoder lets it correct a slip, and it raises an alarm if it can't keep up. It's like telling someone "hold your hand exactly here." They will, however hard you push, until they give up.

- It holds a hanging load firmly and simply. Steppers are strongest at low speed, which is all this rig needs.
- Setup is wiring and a few switches. There is almost nothing to tune.
- It never says how hard it's pulling. A tight line and a slack line look the same to it. So we add a load cell in each line to measure tension.
- It is cheaper, and brake versions exist.
- It hums or whines, more or less depending on the drive.

ODrive (brushless servo): "be here," "move at this speed," or "pull this hard." The motor is a brushless motor, like those in drones and robots. The ODrive board controls the current in its coils directly, many thousands of times a second, and takes commands over CAN. Current is torque, and torque on a drum is line tension. So the ODrive can be told "pull this line with 40 N." If something pulls harder, it gives way. If the line goes loose, it takes up the slack. It's like telling someone "keep this rope tight with this much force." Their hand goes wherever it must.

- It can control tension directly, and it reports current, which gives a rough tension reading without a load cell.
- It can act like a stiff spring around a target length: hold position, but give a little. That is exactly what four lines on a three-axis rig need (below).
- It is very smooth and nearly silent.
- It needs tuning: control gains, and an encoder calibration. Some setups do that calibration by turning the motor at power-up, which a hanging head can't allow. We would need an absolute encoder or a saved calibration.
- It costs about twice as much, and needs a separate brake, a custom motor mount and drum. Stock has been patchy.

Why the difference matters here. Four lines hold a head that has three axes, so one line is always extra. If all four lines are held to exact lengths, any small error (a pulley a few millimetres off, a line that stretched) makes one line go slack or makes the motors fight. Something must manage tension:

With	How tension is managed
Closed-loop steppers	Load cells measure each line. The firmware changes the line lengths a little, many times a second, until the tensions match what the model expects. All in our software.
ODrive	The drives do much of it: each line acts like a spring, or one line is run in "pull this hard" mode. Load cells become a check, not the main control.

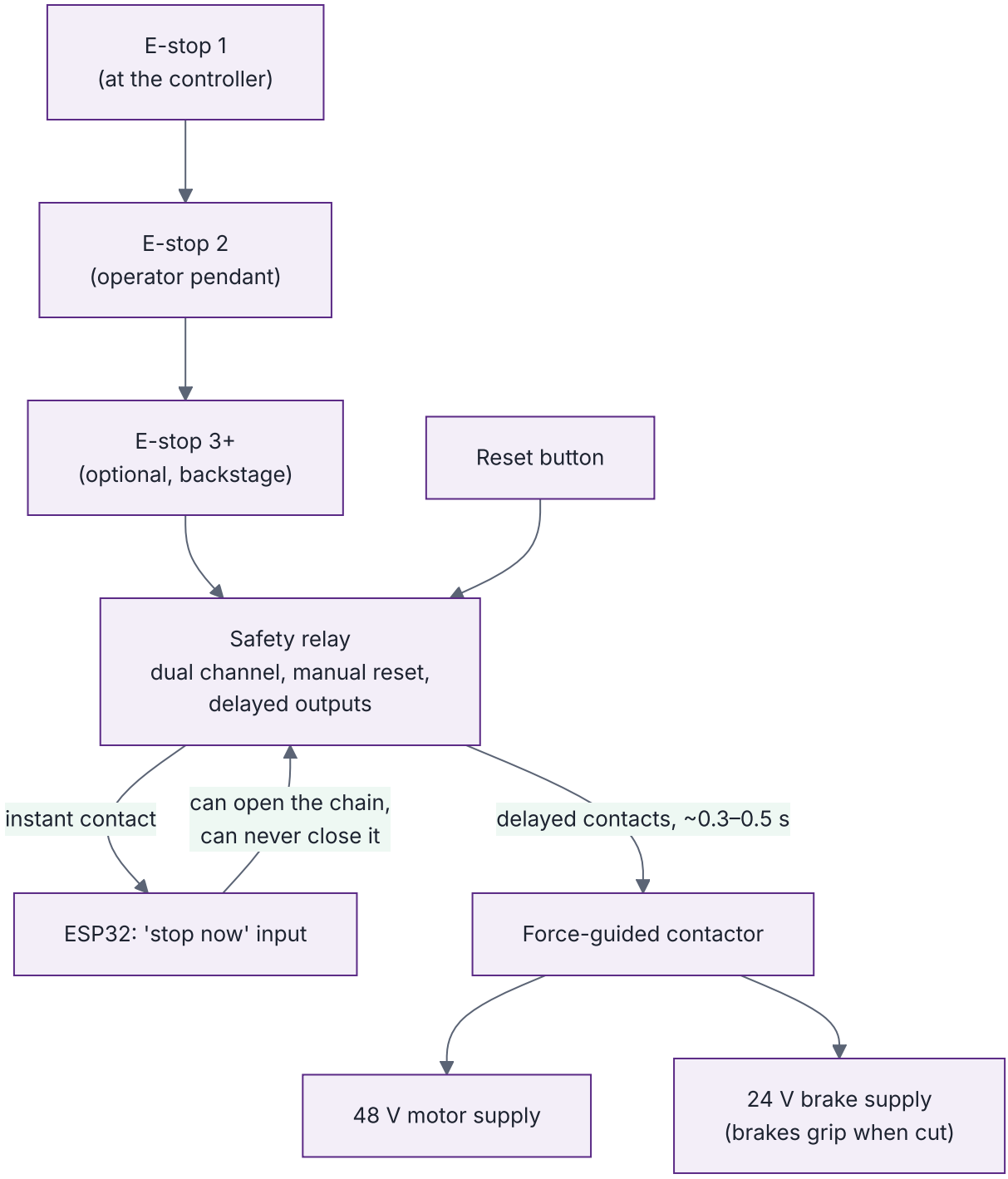
Both work. Steppers put the tension work in our firmware but are simple to start with. The ODrive puts it in the drive, at the price of tuning and cost.

Winch mechanics:

- A **grooved drum with a single layer of line** (a helical groove, about 1 mm pitch). The radius never changes, so motor angle maps exactly to line length. An 8 m line on a 30 mm drum is about 85 turns, about 85 mm of drum.
- A fixed **fairlead** (guide roller) a short way from the drum. It keeps the line's path fixed as it moves along the groove.
- A **tension sensor in each line near the winch**: three rollers in a row, with the middle one on a small beam load cell. Line tension is proportional to the reading, because the line angle over the rollers never changes. Amplifier: HX711 at 80 samples/s to start, or ADS1232 / NAU7802 for faster readings.
- Top pulleys: small sealed-bearing sheaves on swivels, black. The line leaves the pulley at a point that moves a little as the head moves. The kinematics model includes that (section 5.3).

3.3 Safety hardware

The E-stop works in hardware. Software only reports it.



- **E-stop buttons:** at least two. They have red mushroom heads on yellow bases, twist to release, and two normally closed contacts each, wired in series. One sits on the controller enclosure. One is on a pendant with a long cable so the operator can stand where they see the head. Add one backstage if crew work there.
- **Safety relay:** a dual-channel E-stop relay with manual reset and delayed outputs (Pilz PNOZ, Omron G9SE, Phoenix PSR families, among others). The delay gives a **controlled stop first, then power off** (IEC 60204-1 "category 1"). A hanging head shouldn't be snapped to a halt by four brakes at full speed. The ESP32 gets an instant signal to decelerate. If it doesn't, the delayed contacts remove power regardless.

- **Reset** is a physical button. A software command can never reset it, which matches SCMP's "only a physical reset closes the chain".
- **Local/Remote key switch** (recommended, in addition to the touchscreen toggle). Turning the key to Local locks out network control, whatever the screen says. The person working on the rig keeps the key.
- **Stack light or beacon:** red = E-stop, amber = armed, flashing amber = moving, green = safe/disarmed. The crew can see the state in the dark without the screen. Sound a short horn on arm.

3.4 Power and enclosure

- 48 V DC supply for the drives (Mean Well, about 480–600 W), 24 V for brakes, relays and I/O, and 5 V for the ESP32 from the 24 V.
- An IP54 or better enclosure with the touchscreen in the door, behind a clear cover if rain can reach it. Glands or panel connectors for each winch. Feed it from a GFCI outlet.
- Drives in the main enclosure. Run motor, encoder and brake cables out to each winch (screened, labelled, locking connectors). Check the encoder cable length limit. If the posts are far apart, put each drive in a small box at its winch and send it differential step/dir (or CAN with Option B).

3.5 Wind sensor

- An RS-485 Modbus anemometer at the head's height, at the open side of the tent. A cup type (for example DFRobot SEN0483, under $100) is fine to start. Many cheap ones don't report gusts, so the firmware computes gusts itself from fast readings.
- An ultrasonic sensor (no moving parts, reports gusts) is the upgrade: ~$300–900.
- Tension sensing (3.2) is the second, more direct wind sensor. It measures what actually reaches the head. See section 6.

3.6 Rough budget (to check before ordering)

Part	Rough cost
ESP32-P4 board + 7" touchscreen	$60–100
4 closed-loop stepper + drive + brake sets	$600–1,000
Power supplies (48 V, 24 V)	$120–200
Safety relay, contactor, 2–3 E-stops, reset, key switch, stack light	$300–600
4 load cells + amplifiers, I/O carrier board	$150–300
Anemometer	$50–900
Drums, pulleys, bearings, fairleads, mounts, line	$200–400
Enclosure, cable, connectors	$200–400
Frame (rented truss or built)	$300–1,500+
Total	**about $2,000–5,000**

4. The numbers that drive the design

A quick static model (to become a proper sizing tool in Phase 0). It uses the real tent: pulleys at 2.44 m, set 8 cm in from each corner of the 8 ft square. The ball is 0.46 m (18 in) across and rests on the pedestal at 1.0 m. The four lines attach to a ring 0.3 m across on top of the ball.

The space it can fly in

Limit	Value
Ball center when docked	about 1.23 m
Highest ball center (top of ball 0.3 m below the pulleys)	about 1.91 m
Room to rise	**about 0.68 m**
Sideways, each way from center (0.3 m clear of the curtains)	**about ±0.69 m**

Once clear of the table, the ball can also come down beside it. Tensions climb fast as the top of the ball nears the pulleys, so the top limit is firm.

What the head weighs

An 18 in acrylic sphere weighs:

Wall thickness	Sphere weight
2 mm	1.6 kg
3 mm	2.3 kg
4.5 mm	3.5 kg
6 mm	4.7 kg

Add the head form, projector, battery, LEDs and the attachment ring. Somewhere around **4–8 kg** is likely.

Line tensions by head weight

Tension scales with weight. Over the whole workspace, including gentle acceleration (0.5 m/s²), the worst line sees about **14 N per kg of head**. Resting just off the dock, each line carries about 4 N per kg.

Head	Each line, just off the dock	Worst line, anywhere	Line rating at 8:1	Torque on a 12 mm drum
3 kg	13 N	41 N	~75 lb	0.5 N·m
5 kg	21 N	69 N	~125 lb	0.8 N·m
8 kg	34 N	110 N	~200 lb	1.3 N·m
10 kg	42 N	138 N	~250 lb	1.7 N·m

Is heavier more stable?

Yes, against things that push from outside. But only up to a point, and not against everything. A second model treats the ball as a rigid body on four stretchy lines (black braid, ball center at 1.6 m). It pushes the ball with a 22 mph (10 m/s) wind from the open front, which is 4.6 N on an 18 in ball.

Head	Ring 0.3 m across, weight 8 cm low: ball moves	Tilts	Twists, per small nudge*
3 kg	4.8 mm	2.1°	0.27°
5 kg	3.3 mm	1.3°	0.16°
8 kg	2.4 mm	0.8°	0.10°
12 kg	1.9 mm	0.5°	0.07°
5 kg, but ring 0.16 m across, weight at the center	10.7 mm	3.1°	0.28°

* A 0.05 N·m twisting push, such as an off-center gust.

What this shows:

1. **Weight helps against outside pushes:** wind, a curtain brushing a line, a tug. Doubling the weight cuts the movement by about a third to a half. The returns shrink: 3 → 5 kg is a big step, while 8 → 12 kg barely shows.
2. **Weight doesn't help with the ball's own moves.** When the ball speeds up or slows down, the force that makes it rock grows with its weight, and so does the force that steadies it. They cancel out. The rocking period (about 1.3–2 times a second) hardly changes with weight either. What helps here is smooth S-curve moves (section 5.3).
3. **Shape matters more than weight, and it's free.** A low center of weight (battery at the bottom) always helps. Spreading the attachment points wider cuts sway and twist too, but it makes the ball lean when it's off center. See the next section: the points should be about 10 cm apart, not 30 cm.
4. **Weight has real costs:** thicker line (it grows with the square root of the weight, so double the weight is 1.4 times the thickness, and more visible), more to fall if something breaks, and more pull on the frame.
5. **At sane wind levels, none of this is large.** At 10 mph the push is a fifth of the table's, so a 5 kg head moves under a millimetre.

Recommendation: aim for about **5–8 kg** from the parts themselves. Don't add ballast for stability: get it from a low battery, the right attachment spacing and smooth moves first. Build the winches for 10 kg so the weight doesn't hold up ordering. Choose the line once the head is weighed.

Keeping the face towards the audience

A third model finds where the ball actually settles, given the four line lengths and gravity. Two results:

- **Turning is not a problem.** Wherever the ball goes in the workspace, it stays square to the audience to within 0.02°. A line 2 mm too long doesn't turn it measurably either. The ball has no reason to turn: the layout is symmetrical, a round ball catches no twisting wind, and when it does turn, the attachment points swing inwards and every line gets longer, so gravity pulls it back.
- **Leaning is the real effect.** Off center, the lines on the near side get steeper and the far ones shallower. Their pull no longer balances around the ball, so it leans. How much depends on how far apart the attachment points are:

Attachment points apart	Lean, 0.5 m to one side	Lean, far corner	Effort to turn the face 1°
One point	0°	0°	none: it spins freely
6 cm	1.6°	3.1°	0.07 N·m
10 cm	**2.6°**	**5.2°**	**0.11 N·m**

Attachment points apart	Lean, 0.5 m to one side	Lean, far corner	Effort to turn the face 1°
16 cm	4.0°	8.2°	0.18 N·m
30 cm	7.3°	15.1°	0.31 N·m

(5 kg head, center of weight 8 cm below the ball's center. Moving the weight to 15 cm below cuts the lean by about a sixth.)

Spacing the points wider holds the face more firmly but makes the ball lean more. **About 10 cm apart is the proposed balance.** A slight lean towards the side it drifts to may even look natural. The real twisting pushes are tiny, so 0.11 N·m per degree holds the face to a fraction of a degree.

The firmware must model the lean. If it worked out line lengths as if the ball stayed level, the ball would land up to about 2 cm from where it was sent (section 5.3).

Inside the ball: a central spine

The lines pull up to 140 N on the top fitting. Point loads like that can crack acrylic, so the sphere shouldn't carry them on its own. Proposed:

- **A small top fitting**, black, about 10–12 cm across, holding the four line points. It clamps through a hole at the top of the sphere.
- **A spine** runs down from it, hidden in the wig, into the head. The head, projector, mirror and battery all hang from the spine. The sphere is clamped to the same fitting, so it adds no load path of its own.
- **The battery goes at the bottom,** in or below the neck. It is the low center of weight.
- **A small base fitting** at the bottom of the sphere carries the charging pads or coil (below) and seats the ball on the dock. A crystal ball on a stand looks right, so this can be part of the look.

Further points from the model:

- **Four lines, three axes: one line too many.** With all four lines set by position, a few millimetres of error in anchor positions or line length either makes one line go slack or makes them fight each other. The lines must be **tension-managed**, not just position-controlled. Maslow 4 (four belts, ESP32) and Hangprinter (four lines) both ran into this. Section 3.2 explains how the two motor options handle it. Section 5.3 has the firmware side.
- **The posts take an inward pull** of up to about 140 N each (10 kg head), sideways at the top. The frame needs diagonal braces or a top rectangle that ties the posts together.
- **Low-stretch line matters.** With nylon monofilament the ball moves about twice as far in the same wind, and it bounces at a lower frequency.

Wind on the ball

A smooth 18 in sphere (drag coefficient about 0.47) catches:

Wind	Force on the ball	Share of a 5 kg head's weight
4.5 m/s (10 mph)	0.9 N	2%
7 m/s (16 mph)	2.3 N	5%
10 m/s (22 mph)	4.6 N	9%

The lines won't go slack at any of these. The real limit is how it looks: the ball sways and rocks, the curtains billow towards it, and the posts move. Wind inside a tent closed on three sides is much lower than outside, except when it blows straight in the open front. Section 6 turns this into rules.

Power on board

Rough power budget (to firm up once the projector is chosen):

Load	Power
Raspberry Pi Zero 2 W playing video	about 1.5–2.5 W
Small projector	about 2–10 W, by model and brightness
LEDs	about 1–3 W
Total	**about 5–15 W**

A full-evening battery is light. 4 hours at 15 W is 60 Wh. Add 25% for battery ageing and cold nights: about 75 Wh. A lithium-ion pack stores very roughly 150 Wh per kg, so that's about **0.5 kg**. A pack for 2 hours would save only about 0.25 kg. The ball should weigh 5–8 kg anyway, and weight at the bottom is the good kind. So a smaller pack buys almost nothing.

Recommendation: carry the full 4-hour pack, and charge on the dock as well. The dock then tops the battery up between shows and charges it overnight. Nobody has to open the ball, and a long night or a cold one can't run it flat. The dock is a convenience, not something the show depends on.

Charging on the dock, two ways:

Way	For	Against
Spring-loaded contacts (gold-plated pogo pins on the pedestal, pads on the ball's base)	Simple. High power, so a fast top-up.	Exposed metal outdoors: dew and dirt. The pads must line up.
Inductive (a Qi-style coil in the pedestal, another in the ball's base)	Nothing exposed. Works through the acrylic. Sealed.	Common modules give 5–15 W, so it's slower. The coils must be within a few millimetres of each other. The dock's seat lines them up.

Either is fine with a full-evening pack. Inductive suits a damp outdoor tent better.

The controller switches dock power on only while the ball is seated and disarmed, and it reports the ball's charge if the Pi sends it.

Power path matters. Many power banks cut their output for a moment when a charger is connected or disconnected. That would reboot the Pi every time the ball lands or lifts off. Use a battery board with proper power-path handling (a "UPS"-style board for the Pi, or a pack whose output never drops while charging). Test it by docking and lifting while video plays.

Other points:

- A projector in a closed sphere gets hot. Plan hidden vents, or a projector small enough to run cool.
- Projector candidates to evaluate: TI's DLPDLCR2000EVM (a tiny DLP projector board made for a Raspberry Pi, 640×360, up to 30 lumens) and its bigger sibling DLPDLCR230NPEVM (1080p, 100 lumens). Check that they work with a Zero 2 W, and check their throw and power. In a dark scene, on a small face, 30 lumens may be enough.

- The Pi Zero 2 W can be a **Show Controller follower** (design section 9). It plays its own copy of the media, in sync with the show, and takes cues over Wi-Fi. If Wi-Fi drops, it carries on by itself. Motion never depends on this link: the motion controller stays on Ethernet.

5. Software

5.1 Build on what, exactly

The open-source CNC and cable-robot projects were reviewed. None fits as a base, but several are good references:

Project	What to take from it	Why not use it as is
Hangprinter (and its RepRapFirmware and Klipper kinematics)	Cable kinematics for "translational cable robots with a central effector", including Spidercam-style layouts. Line flex compensation, pre-tension handling, ODrive torque control.	It's a G-code printer stack. It doesn't run on an ESP32 with SCMP.
Klipper `winch` kinematics	The basic per-line length math.	Basic, with no homing or limits. It runs on a Linux host.
Maslow 4 (ESP32-S3, FluidNC fork)	Four-line tension and calibration on an ESP32. Their forum is a catalog of what goes wrong with over-defined line systems.	Built for a CNC router and G-code.
FluidNC / grblHAL	Config, step generation and planner ideas. grblHAL has (experimental) wall-plotter kinematics.	G-code motion planning doesn't map onto SCMP's per-axis moves, timed moves, jog and heartbeat. We also need our own tension, wind and safety logic.
FastAccelStepper	Step pulses from hardware (RMT). It lists the ESP32-P4 at 200 kHz on RMT, marked experimental.	Use as a library, or as a reference for our own RMT driver.

Recommendation: purpose-built firmware on ESP-IDF (5.3 or later, required for the P4), in C++, with FreeRTOS tasks. LVGL 9 draws the UI, through Espressif's esp_lvgl_port. Lay out the screens with EEZ Studio or SquareLine Studio.

5.2 Tasks

Task	Core	Rate	Does
Motion	1 (alone)	1 kHz	Runs the per-axis motion profiles, does the kinematics (5.3), trims tension, and sends step targets. Checks every safety limit each cycle.
Supervisor	1	100 Hz	The state machine (5.5). Watches E-stop, heartbeat, drive alarms, tensions, wind and the watchdog pin.
SCMP server	0	on demand	UDP and TCP on port 7720. Handles `seq` duplicates, acks and status (5.4).
Sensors	0	80–100 Hz	Load cells, anemometer (Modbus), inputs.
UI	0	30 fps	LVGL screens (5.6).
Config and log	0	on demand	Calibration in flash (NVS), an event log in RAM and on flash.

The motion task never waits on the network, the screen or flash.

5.3 Kinematics and tension

- **Inverse kinematics** (every 1 ms): for head position p, each line's length is the distance from its pulley exit point to its attachment point on the head, plus the arc wrapped on the pulley. Motor angle = length ÷ drum radius.
- **The lean is part of the kinematics.** The ball leans a few degrees off center (section 4), which moves the attachment points. Working on a PC, a model finds the ball's resting pose and line lengths across the workspace. Those go into a table that the controller looks up and interpolates. Without it, the ball would land up to about 2 cm off target, and the tension model would be wrong.
- **Tension management:** a slow outer loop (10–50 Hz) compares measured tensions with the tensions the force model predicts. It then lengthens or shortens each line by a small amount to match. Large errors are faults (below).
- **Flex compensation** (later): length correction for line stretch under load, as in Hangprinter. Small with UHMWPE line, large with nylon.
- **Workspace check:** each target, and each point on the way to it, must be inside the configured box, outside keep-out zones (the dock, set pieces, curtains), and in a place where every line can stay tight. Targets outside are refused with `out_of_range`, never clamped (SCMP rule 5).
- **Profiles:** S-curve (jerk-limited) per axis. Smooth starts and stops matter more than speed here. They keep the head from bouncing on its lines.

Line faults (all stop the rig):

Fault	When
`slack_line`	A line's tension stays below its minimum (say 5 N) for 100 ms.
`overload`	A line's tension goes above its maximum (say 150 N).
`tension_mismatch`	The measured tension pattern doesn't match the model. This catches bad calibration, a snag on a curtain, or a broken line.
`drive_alarm`	A drive reports a position error.

5.4 SCMP: how the rig looks to Show Controller

SCMP is per-axis, so the controller presents **Cartesian axes** rather than motors. Show Controller never sees lines or motors.

Axis	Units	Meaning
`x`	mm	Left (−) to right (+), as the audience sees it. 0 is center.
`y`	mm	Depth. 0 is the show plane, + is upstage. **Missing in 2-axis mode.**
`z`	mm	Height above the floor.

This works with SCMP version 1 as written:

- `move` on each axis runs that axis's own profile. Three `move`s with the **same `duration`**, sent together, give a coordinated, straight-line move. Show Controller's timelines already do this with move items at the same time.
- `jog` works per axis, so the joystick on Show Controller's Motion page flies the head like Skycam's operator does.

- Named positions ("Leota hidden", "Leota center") live in Show Controller, as design Q15 says. Calibration (pulley positions, drum radius, workspace, tension limits, wind thresholds) lives in the controller, as `config` keys.

How the axes share state, since they're one machine:

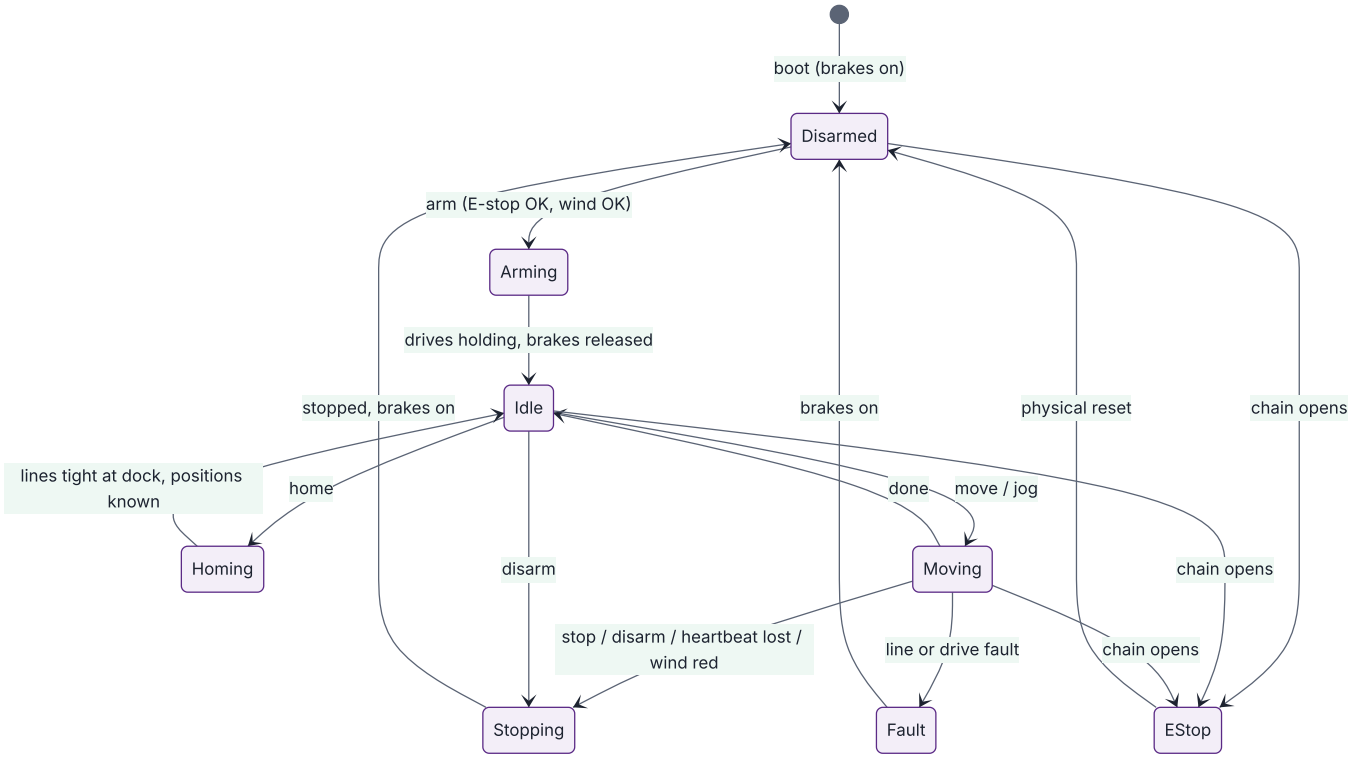
- `arm`, `disarm`, `home` and `stop` act on the **whole rig**, whatever `axis` says. Every axis then reports the same `armed` and `homed`. One axis can't move alone while another is disarmed.
- Disarmed means the brakes are on. Arming enables the drives, waits for them to hold, then releases the brakes. Disarming does it in reverse after a controlled stop.

Proposed SCMP additions (version 1.1, to agree in the BEDShowController repo):

Addition	Why
Error `local`	The controller is in Local mode (touchscreen toggle or key). Commands that move it, arm it, home it or change config are refused. `hello`, `heartbeat`, `status` and `stop` are always accepted.
Error `weather`	Wind lockout (section 6).
`status` fields: `"remote": true/false`, `"wind": {"speed", "gust", "level"}`, `"lines": [{"tension", "length"}]`	Show Controller can show why the rig won't arm, and log conditions.
Faults `slack_line`, `overload`, `tension_mismatch`, `drive_alarm`, `wind`	New fault causes, from section 5.3 and section 6.
Optional: one `move` with several axes (`"axes": {"x": …, "z": …}`)	Starts them at exactly the same instant, without relying on three packets landing together. Not needed on a wired LAN, but cleaner.

Before building this, check that Show Controller's SCMP client ignores status fields it doesn't know.

5.5 States



- Boot is always **Disarmed** and **not homed**. Incremental encoders don't remember position, and a line may have been moved by hand.
- **Heartbeat loss** only applies in Remote mode. In Local mode the operator at the screen is in control.
- **Homing** uses the **dock**: the pedestal on the table, at a known spot, where the ball rests with its lines loose. A shaped seat (a ring or cup) puts the ball in the same place every time. Homing reels each line in slowly until its tension reaches the pre-tension (say 10 N). The rig now knows every line length from the dock's position. It lifts 30 mm and checks that the tensions match the model. This is better than limit switches, which a cable robot doesn't really have. The dock is also where the ball charges, and where it parks overnight and in bad weather.
- **The table is a keep-out zone.** The ball may not come lower than the table top while it is over the table, except straight down onto the dock.

5.6 Touchscreen

7" landscape. Big targets, readable from a step back. Dark theme for a dark tent.

Screen	Contents
Status (home)	A large state banner (E-STOP, DISARMED, ARMED, MOVING, FAULT). Remote: ON/OFF. Show Controller link: connected, from which address, last heartbeat. Position x/y/z. Homed. Wind: now, gust, level (green/amber/red). Four tension bars. Active faults.
Controls	Remote enable/disable. Arm and disarm (press and hold 1 s). Home. Park at dock. A large STOP. Clear fault.
Jog	Hold-to-run XY pad and Z slider, with a speed limit. Go to dock or the service position. Local mode only.

Screen	Contents
Diagnostics	Each winch: position, drive alarm, brake, tension. Every input (E-stop chain, reset, key). Network settings and counters.
Setup (PIN)	Pulley positions, drum radius, workspace box, keep-out zones, tension limits, wind thresholds, 2- or 3-axis mode. Calibration wizard.
Log	Recent events in plain sentences.

The touchscreen STOP is a convenience, like Show Controller's stop-all. The physical E-stops are the safety function.

5.7 Testing without hardware

- Build the kinematics, profiles and state machine as plain C++ with no ESP-IDF dependencies. Unit-test them on a PC.
- **A simulator that speaks SCMP:** the same core code built for Linux, or a cable-robot mode added to Show Controller's Python simulator. It shows the head and lines in 3D and tensions per line. Timelines can be written and checked before the frame exists.
- Run Show Controller's own SCMP client and motion module against the firmware on the bench to prove protocol compatibility.

6. Wind and weather rules (proposed)

Three levels. The firmware enforces them. Show Controller and the screen both show them.

Level	When (starting values, to set in testing)	The rig
Green	3 s gust under 7 m/s (15 mph), and tensions steady	Full workspace and speed.
Amber	Gust 7–11 m/s (15–25 mph), or tensions start to swing	Speed limited to half. Workspace shrinks to the low center, away from curtains and corners. Warns Show Controller and the screen.
Red	Gust over 11 m/s (25 mph), **or** wind sensor fault, **or** tensions swing past the red limit	Flies to the dock, disarms, and refuses `arm` (`weather`) until conditions have stayed Green for 15 minutes. A person must re-arm it.

- **Gust** = the highest 3-second average in the last 10 minutes. **Steady** = 2-minute average. Both are computed in firmware from 1 Hz or faster readings.
- **Tension swing** = how much each line's tension varies over a few seconds while the head stands still. It measures what actually reaches the head, including what the anemometer misses (a gust through the curtains, a billowing wall). It can raise the level by itself.
- A **missing or stuck sensor counts as Red.** Fail safe.
- These limits are for the prop. The tent has its own wind rating from its maker and local permits. If that is lower, it wins, and the tent should be cleared first anyway.
- Separate from wind: stop for lightning nearby (crew policy). Inspect lines after rain. Nylon line changes length when wet, another reason to prefer UHMWPE.

How to set the real numbers (Phase 5): put the anemometer next to the head. Blow on it with a fan or leaf blower at rising speeds, at the workspace corners. Find the speed where (a) a line drops below its minimum, (b) the head visibly moves, (c) the curtains reach the lines. Set Amber below the lowest of those, and Red at it.

Site rules that make the numbers better:

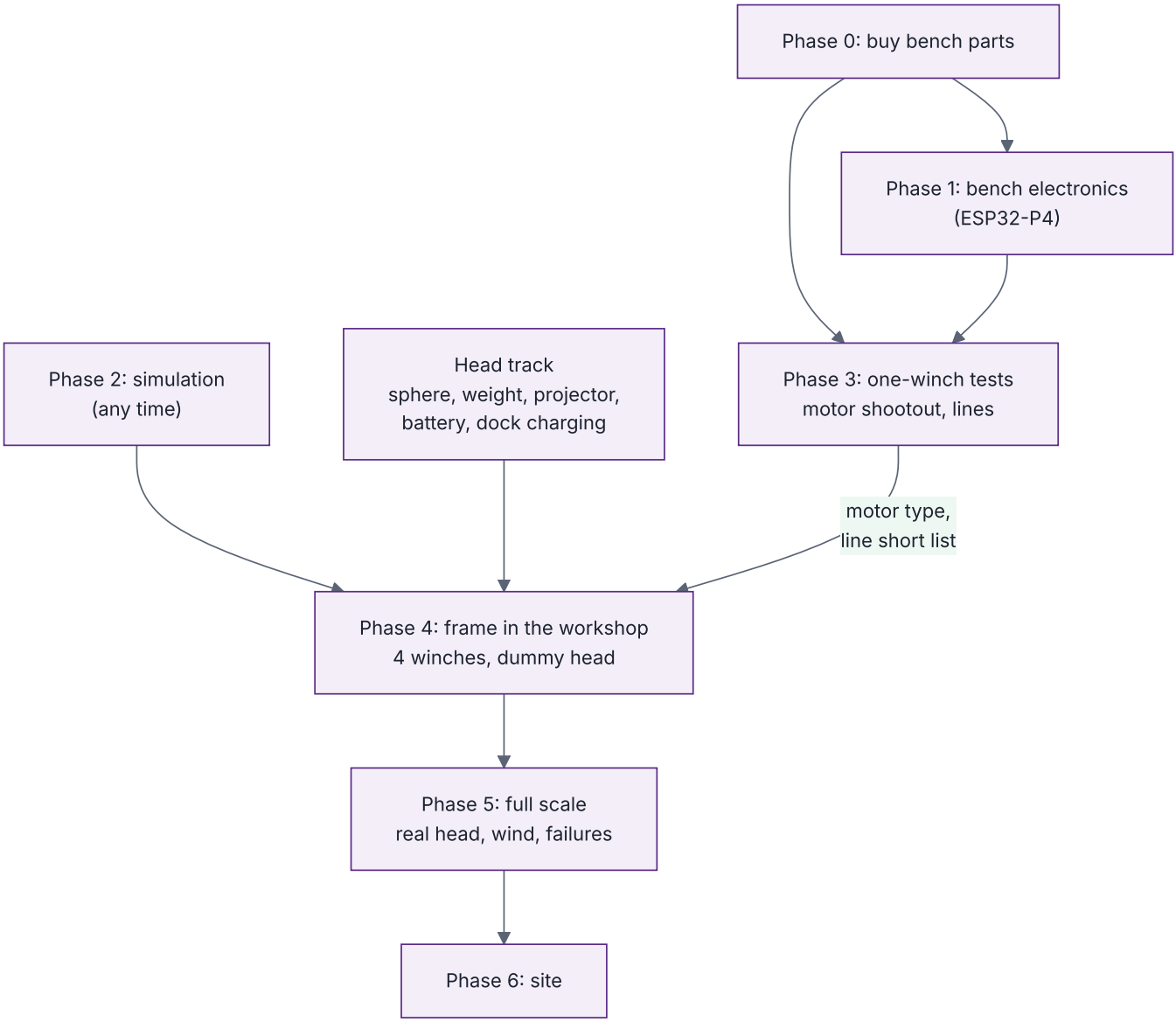
- Face the tent's open side away from the usual wind, if you can.
- Tie and weight the curtain bottoms. Keep at least 0.3 m between the curtains and the ball's path. The tent is small, so every bit of curtain movement matters.
- No guests under the head or in reach of a line. Barricade the area under the workspace.

7. Risks and what handles them

Failure	What happens	Covered by
Line breaks	Head swings on 3 lines or drops	Design factor on line. Inspection and replacement schedule. `slack_line` stops the rig. Nobody under the head.
Power cut or E-stop	Drives lose power	Fail-safe brakes hold the head where it is.
ESP32 hangs	No steps, no heartbeat answer	Watchdog charge pump drops drive enable. Brakes on. ESP32 hardware watchdog reboots it.
Network lost	No heartbeat	Stops and disarms after 0.5 s (SCMP Q8).
Motor stalls or skips	One line wrong length	Drive alarm, and tension mismatch.
Line snags a curtain or set piece	Tensions jump	`overload` or `tension_mismatch`. Keep-out zones. Curtain clearance.
Frame sways	Head moves with the posts	Freestanding, ballasted frame. Tension swing raises the wind level.
Calibration drift (line creep, temperature)	Slowly growing tension mismatch	Re-home at the dock at the start of each show day. Tension trim. Low-stretch line.
Wrong command from Show Controller	Unexpected move	Soft limits, workspace checks, Local mode, and SCMP's rules. Arming by schedule only if the device allows it (Q7).

8. Plan of work: tests that settle the design

Two tracks run side by side. The **motion track** doesn't need the head to be finished: the winches are sized for the heaviest likely head (10 kg), and dead weights stand in for it. The **head track** settles the payload: weight, center of weight, projector and power. They meet in Phase 4, on the full frame.



Motion track

Phase	Tests	Settles
0. Buy bench parts	See "What to buy now" below.	

Phase	Tests	Settles
1. Bench electronics	Ethernet and an SCMP session with Show Controller's SCMP client. Touchscreen running. **Step pulses from the P4** on a logic analyzer: smooth at very slow speeds, no glitches while the screen and network are busy. A CAN link to the ODrive. One E-stop that cuts the 48 V supply through a contactor.	That one ESP32-P4 can do motion, network and screen together. Step pulses on the P4 are the least proven part (FastAccelStepper lists P4 support as experimental), so this test comes first.
2. Simulation	Kinematics with the lean (section 4), profiles and state machine, unit-tested on a PC. An SCMP simulator.	Firmware structure. Needs no hardware.
3. One-winch tests	A winch on the floor, a line up over a pulley at about 2.4 m (a beam or a temporary post), down to a dead weight. Water jugs or sandbags of 2, 5, 10 and 14 kg give line tensions across the full range (20–140 N). Run both motor types on the same rig. See below.	Motor type, drum size, brake size, and a short list of lines.
4. Frame in the workshop	Buy three more winches of the chosen type. Build the real 8 ft frame. Fly a dummy head of the right weight and size (a cheap ball with weights inside, hung from a copy of the top fitting). Check that the lean and facing match the model. Dock homing, tension management, settling after moves, S-curve tuning, the joystick from Show Controller.	The control approach works. The final pin plan and I/O carrier board.
5. Full scale	The real head. Wind tests with a fan or leaf blower (section 6). Failure tests: E-stop at full speed, unplug Ethernet, cut power, a line break on the dummy head.	Wind limits. Final line choice.
6. Site	Install under the tent. Checklists. A show night run from Show Controller.	Ready for shows.

Phase 3 in detail. The same tests on the closed-loop stepper and the ODrive:

Test	How	Looking for
Noise	A phone sound meter at 1 m, in a quiet room at night. Holding still, crawling (5 mm/s), cruising (0.1–0.3 m/s).	Whether it can be heard where the audience stands.
Crawl smoothness	Slow-motion phone video of the weight, or a small accelerometer taped to it, at 2–10 mm/s.	Visible stepping or shiver. A floating head creeps slowly, so this matters most.
Holding	14 kg held for an hour. Motor temperature, position drift.	Heat and drift through a long show.
Brake	Cut power while moving down with 10 kg. Measure the drop and the jolt.	That the brake holds, and the stop isn't too harsh.
Tension reading	Load cell against known weights. On the ODrive, motor current against the load cell.	Whether the load cells are needed for control, or only as a check.
Faults	Stall the motor on purpose. Pull the step signal. Drop the CAN link.	The drive raises an alarm, and the controller sees it.
Lines	Candidate lines on the same rig: black braid at 100, 150 and 250 lb, clear monofilament, fluorocarbon. Stretch under load. Creep with 10 kg left hanging overnight. Length change when wet. 1,000 trips over the pulley and drum.	Stretch, creep and wear.

Test	How	Looking for
Visibility	The lines hung against black curtain, at night, under the planned lighting, viewed from where the audience stands. Photographs from the audience's position.	Which lines disappear. The final size waits for the head's weight.

Head track (runs in parallel)

Step	Work	Settles
H1. The sphere	Buy the 18 in acrylic sphere early. It is the biggest single weight and sets the fittings. Weigh it, measure the wall, and check the hole at the top.	Sphere weight, and the top fitting design.
H2. Projector	Pi Zero 2 W, candidate projectors, a mirror and a face blank on the bench. Brightness, throw, heat, power draw.	The projector, and the power budget.
H3. Power	Battery and power-path board. Measure a 4-hour run. Dock charging (contacts or inductive), docking and lifting while video plays.	Battery size, charging method, no reboots on docking.
H4. Weigh it	The full head on the spine, with wig, projector and battery. Weight, and where the center of weight is.	Payload: the final line size, and the lean model's real numbers.

What to buy now

The controller and a winch of each type don't depend on the head's final weight, so they can be bought now. About **$800–1,100** in all:

Part	About	Notes
Espressif ESP32-P4-Function-EV-Board, with the 7" LCD	$55–90	The LCD is optional in the package, so order the kit that includes it. 26 GPIOs on a 40-pin header: plenty for the bench.
Closed-loop stepper, NEMA 23 about 3 N·m, with 24 V brake, and its drive	$150–250	Option A. Check the brake holds at least 2 N·m.
ODrive S1 + M8325s kit	$280–340	Option B. The brake comes later, if B wins.
48 V supply (about 400 W), 24 V supply	$100–150	
ODrive brake resistor	$10–20	Lowering a weight turns the motor into a generator. The energy has to go somewhere, or the supply voltage jumps.
Beam load cell (20–50 kg) and HX711 amplifier	$15–30	
CAN transceiver module (3.3 V, e.g. SN65HVD230)	$5–10	The P4 has CAN controllers but no transceiver.
Level shifter or 26LS31 line driver for step/dir	$5–10	The P4's 3.3 V outputs may be too weak for the drive's opto inputs.
USB logic analyzer	$15–25	To see the step pulses.
Pulleys, bearings, a drum (3D printed is fine for testing), line samples	$60–120	
One E-stop button and a contactor	$40–80	A test-bench version of the safety chain.

Wait on: the other three winches (until Phase 3 picks a type), the safety relay and final E-stops, the anemometer, frame material, and the final line size.

Bench safety: a 14 kg weight on a test line is dangerous. Hang it over a padded area that nobody stands under. Add a second, slack safety line to the weight. Keep the E-stop in reach. Test the ODrive without a brake only with light weights, or with the weight resting on a catch.

9. Open questions

#	Question	Why it matters
Q1	Answered: 8 ft tent, pulleys at 8 ft, pedestal at 1.0 m.	Gives 0.68 m of rise and ±0.69 m sideways (section 4).
Q2	Ball about 18 in across; face always to the audience (answered). **What will the whole head weigh?** How thick is the sphere's wall?	Winches are sized for 10 kg. The line is chosen once the head is weighed (section 4).
Q3	Pi Zero 2 W, projector in the neck with a mirror, Li-ion battery (answered). **Which projector?** Contacts or inductive charging on the dock?	Power budget, heat, and the dock design (section 4).
Q4	3-axis from the start, or 2-axis first and 3-axis later?	Same hardware. Only commissioning order changes.
Q5	Option A (closed-loop steppers with brakes) or Option B (ODrive brushless with torque control)?	Cost, tuning effort and how tension is managed. A is the lead choice. Phase 3 tests one of each and decides.
Q6	How fast should it fly at most?	Sets drum size and motor choice. 0.3 m/s is assumed.
Q7	Can guests get near or under the workspace? Where will the operator stand?	Barricades, E-stop count and pendant cable length.
Q8	Design factor for the lines (theatre practice is often 8:1–10:1), and a line replacement schedule?	Line size, and so visibility.
Q9	Are you happy to extend SCMP (section 5.4) in the BEDShowController repo?	`local`, `weather` and the extra status fields need it.
Q10	Budget ceiling?	Picks between the options in sections 3.2 and 3.5.

Sources

- SCMP version 1: `docs/scmp.md` and `docs/design.md` section 11 in ncory/BEDShowController.
- ESP32-P4 boards: Waveshare ESP32-P4 range, Waveshare P4 Module DEV-KIT (Electronics Lab), Espressif ESP32-P4-Function-EV-Board (CNX), NuttX board page (IP101 Ethernet, 7" 1024×600 touch), Olimex ESP32-P4-DevKit, ESP-IDF TWAI on ESP32-P4 (3 controllers), Zephyr: Waveshare ESP32-P4-ETH.
- Step generation: FastAccelStepper.
- Drives and motors: Leadshine closed-loop NEMA 23 with brake (DamenCNC), CL57T V4.1 driver, ODrive S1 + M8325s kit, Teknic ClearPath SDSK.
- Cable robots: Hangprinter, RepRapFirmware Hangprinter kinematics, Hangprinter 2023 blog (line forces), Klipper flex compensation for winch kinematics, Maslow 4 forum, grblHAL config (wall plotter kinematics), Grbl_ESP32 / FluidNC.
- Wind sensors: DFRobot SEN0483 RS-485 wind speed, Delta Ohm HD53LS (reports gusts), PCE-WS US2 ultrasonic.